

Ex 1: For each of the following functions, determine the poles and the residues at the poles:

(a) $\frac{2z+1}{z^2-z-2}$, (b) $\left(\frac{z+1}{z-1}\right)^2$, (c) $\frac{\sin z}{z^2}$

a) $z=2, z=-1$ poles

$$\text{Res}[f, 2] = \lim_{z \rightarrow 2} (z-2) \frac{2z}{(z-2)(z+1)} = \frac{4}{3}$$

$$\text{Res}[f, -1] = \frac{2}{3}$$

b)

$(z-1)^2=0, z=1$

$$\text{Res}[f, 1] = \lim_{z \rightarrow 1} \frac{1}{1!} \cdot \frac{d}{dz} \left\{ (z-1)^2 \frac{(z+1)^2}{(z-1)^2} \right\}$$

$$= \lim_{z \rightarrow 1} 2(z+1)$$

$$= 4$$

c) $z^2=0, z=0$

$$\text{Res}[f, 0] = \lim_{z \rightarrow 0} \frac{1}{1!} \frac{d}{dz} \left\{ \cancel{z} \cdot \frac{\sin z}{z^2} \right\}$$

$$= \lim_{z \rightarrow 0} \cos z$$

$$= 1$$

Ex 2: Find the zeros and poles of $f(z) = \frac{z^2 + 4}{z^3 + 2z^2 + 2z}$ and determine the residues at the poles.

$$\frac{z^2 + 4}{z^3 + 2z^2 + 2z} = \frac{z^2 + 4}{z(z+1+i)(z+1-i)}$$

$$z^2 + 4 = 0$$

$$z^2 = -4$$

$$z^2 = 4i^2$$

$$z = \pm 2i$$

Zeros of $f(z) \Rightarrow 2i, -2i$

poles of $f(z) \Rightarrow z=0, -1-i, -1+i$

Residue,

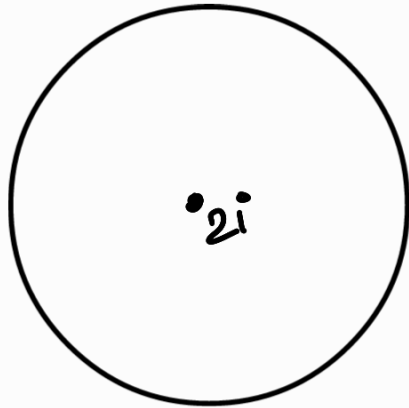
$$\text{Res}[f, 0] = \frac{0^2 + 4}{(0+1+i)(0+1-i)} = \frac{4}{(1+i)(1-i)} = 2$$

similarly for rest 2

Ex 4: Evaluate $\oint_C \frac{2z^2 + 5}{(z+2)^3(z^2+4)z^2} dz$ where C is (a) $|z-2i|=6$, (b) the square with vertices at $1+i$, $2+i$, $2+2i$, $1+2i$.

poles, $z = -2, 2i, -2i, 0$

a)



$$-2 \checkmark$$

$$0 \checkmark$$

$$2i \checkmark$$

$$-2i \checkmark$$

$$|-2-2i|$$

$$= \sqrt{8}$$

$\vdots$

[similar for rest]

$$\text{Res}[f, 2i] = \frac{2 \cdot (2i)^2 + 5}{(2i+2)^3 (2i)^2 (2i+2i)}$$

$$\text{Res}[f, -2i] =$$

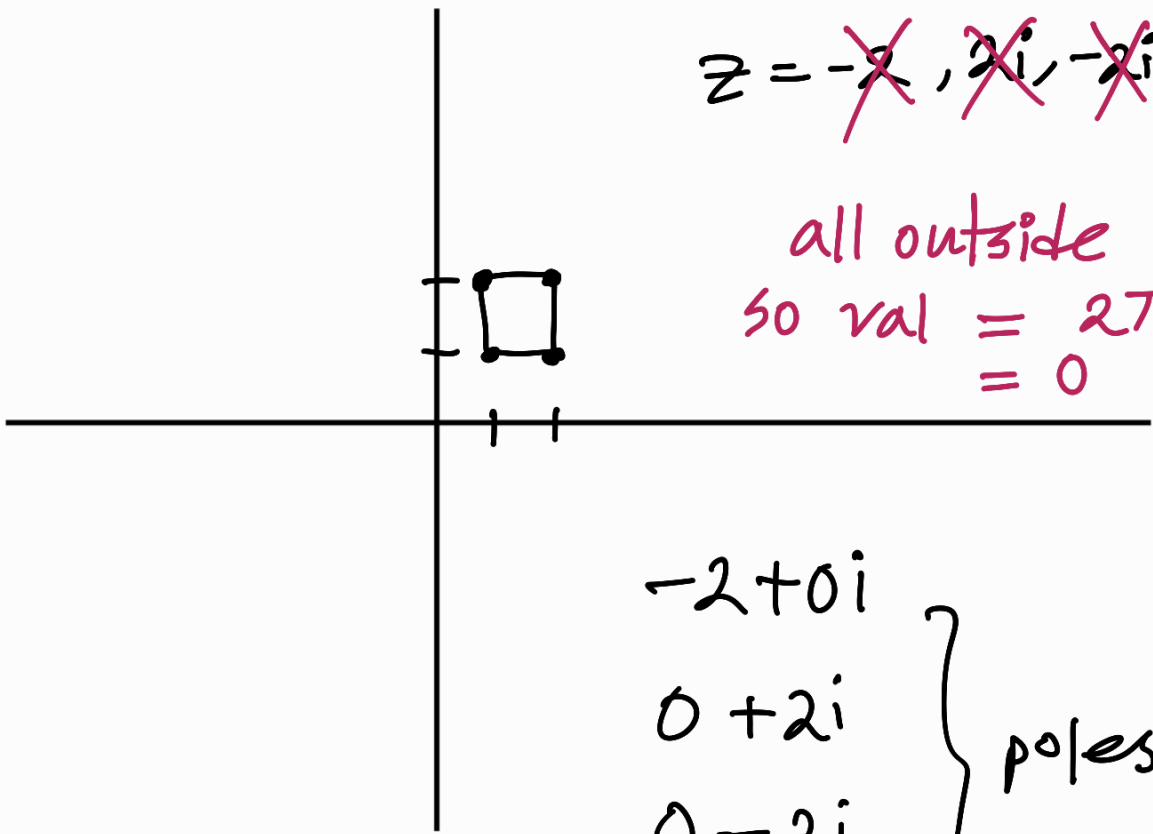
$$\text{Res}[f, 0] = \lim_{z \rightarrow 0} \frac{1}{1!} \cdot \frac{d}{dz} \left\{ \frac{2z^2+5}{(z+2)^3(z-2i)(z+2i)} \right\}$$

$$\text{Res}[f, -2] = \lim_{z \rightarrow -2} \frac{1}{2!} \frac{d^2}{dz^2} \left\{ (z+2)^3 f(z) \right\}$$

b)

$$z = -2, 2i, -2i, 0$$

all outside
so val = $2\pi i \times 0$
= 0



$$\left. \begin{array}{l} -2+0i \\ 0+2i \\ 0-2i \\ 0+0i \end{array} \right\} \text{poles}$$

ex 5,6 similar